Skyrim + Minecraft mod!
MineFriends, a Skyrim mod marries the popular RPG with Minecraft
http://dgit.in/OktYrd

our tests, the cards were pretty powerful. Turns out they are as good as the current CPU super-computers and are used in the field of medicine, chemistry, astrophysics, physics etc.

GPUs have a lot of simple cores which are geared towards simple instruction sets whereas CPUs have 2-8 cores which are general-purpose cores that can compute a wide variety of instructions. GPUs can deliver up to a teraflop of computation at a fraction of the power compared to CPUs while keeping the die size constant.

Anyone can have their own little supercomputer by hooking up a few cards in SLI/CF. Stuffing a mid-range motherboard with a couple of 680s or 7970s can provide immense computing power which small companies can make use of without investing a lot into building a computer cluster.

HOW WE TESTED

RIG:
Motherboard - ASRock X79 Extreme6/GB
Processor - Intel Core i7 3960X
RAM - KHX2133C11D3K4/16GX (Quad Channel)
SSD - Kingston HyperX SH100S3 120GB
SMPS - CoolerMaster SilentPro M850
Monitor - Alienware OptXTM AW2310 3D

This year the focus has shifted to get a better overview of the capabilities of a GPU. So aside from gaming benchmarks we had several OpenGL compute benchmarks. We tried to maintain the parameters the same. Both manufacturers have technologies which are proprietary and provide a lot of computation power if utilised properly, however, it would be unfair to the other manufacturer if these things were focussed upon.

Performance

3D Mark 11
For its focus on DirectX 11 which include tessellation, shaders and multi-threaded CPU tests.

AliensVsPredators
Quite a scalable game which plays well on low-end as well as high-end cards.

Sniper Elite V2
Quite similar to AliensVsPredators except that it is more stressful than AVP and heated up the GPU quite well.

TessMark
Uses the OpenGL 4 API to test tessellation on all supported GPUs. Preset used - “Insane”.

FurMark
Another OpenGL test which can stands true to its label as a “burn-in test”. We achieved the highest temperatures on all cards using this benchmark.

LuxMark
An OpenCL benchmark tool which renders a scene. Unit considered - “Samples/sec”.

AESEncryptDecrypt
Encypts an image using 128bit AES encryption. For the tests we ran 1000 iterations. However, the results though being within a narrow range of values varied greatly across multiple runs, hence we’ve included this just as an indicator and haven’t included them in the score.

Features

By now everybody must be aware that more the memory does not always provide for better gaming performance. BIOS switches are almost a necessity given the cards that come at a high price do offer the capability to scale well and remain stable. These cards tend to come with more power phases and allow for stable overclocks. We did find one card which was a 7970 and didn’t come with a BIOS switch. This is absolutely ridiculous especially when the damn thing cost Rs.10K higher than the other 7970s all of which had a BIOS switch. Custom coolers are always a welcome feature. Copper heat sink + heat pipes + good fans = A really good cooler.

Build quality

The cards were judged on the choice of materials used for the casing of the cards. Reinforcement was also accounted for as it is of prime importance.The big cards tend to warp after quite a while given the high temperatures and the fact that it is kept at a 90 degree angle to the motherboard. Heat pipes were given importance as well. As for overall cooling, judging that is difficult as all other parameters need to be equalised to test cooling performance, so we kept it simple and points were given for those who blew air outside the case rather than for those who kept it circulating inside.

SLI VS CROSSFIREX

Once we were done, it was pretty evident that the scores were in favour of SLI in most of the gaming benchmarks, especially at higher resolutions while CF reigned supreme at lower res. It doesn’t make much sense to spend all this money, get the best possible configurations and then play at low-res to work out the guilt of spending so much. The CF whoops SLI’s behind when it comes to OpenGL/OpenCL computation benchmarks.

3D Mark 11
SLI is ahead in all benchmarks except on the Extreme configuration, but those are within the error margin of 2%.

GAMING

Unigene Heaven 3D
Again the SLI is ahead at high resolutions with a huge lead of 29% while CF takes a meagre lead of 11% at low resolutions.

Resident Evil
The performer in this benchmark seems to be SLI with a lead of 8-11% but the scores varied by a huge margin each time the benchmark was run.

Dirt 3
The results turned out to be similar to Resident Evil except this time it was in the favour of CF by the same amount.

Lost Planet 2
SLI was ahead by upto 9% on high resolutions while the lead reduced at lower resolutions.

Aliens vs Predator
CF takes the lead by quite a wide performance margin as high as 33% over SLI.

Sniper Elite V2
Again CF scooted straight ahead but with a greater lead of 72%.

	CPU		GPU	
Iterations	Encrypt	Decrypt	Encrypt	Decrypt
1	0.185765	0.185286	0.195105	0.201759
100	3.99462	4.00582	0.300444	0.320631
1000	39.1105	40.5503	1.39408	1.29666
10000	393.684	384.493	9.91126	10.8993

That’s the power of your GPU. (GTX 680). Time taken for the operation. (Seconds - Lesser is better)

Thats 126 FPS ahead! Since this is a fairly new benchmark we believe that driver issues once resolved will allow NVIDIA to provide better performance, however, we don’t see it beating CF with the same.

Batman: Arkham City
NVIDIA was the straight winner. What was puzzling was that the NVIDIA logo at the start of the game stuttered like hell even on the flagship Radeon card while even the NVIDIA GT610 which is a rebranded GT520 managed to smoothly render the logo. No doubt the game is optimised for NVIDIA and they’ve made it quite clear. However, this doesn’t mean we can discount SLI’s lead over CF as they did manage a good 16%.